



7049 - A Unique Opportunity to Probe Lensing-Magnified Star Clusters in a Low-Mass Galaxy at $z_{\text{spec}}=8.3$ with JWST

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) Firefly

ABSTRACT

We propose for high signal-to-noise spatially resolved spectroscopic observations of the Firefly Sparkle (FS), a highly magnified galaxy at $z_{\text{spec}} = 8.3$, using NIRSpec/PRISM IFU to obtain spectra of the individual star clusters.

These observations are only possible due to NIRSpec's unparalleled sensitivity, making it uniquely capable of resolving the spatially resolved ISM conditions within and around the star clusters in a lensing magnified low-mass galaxy at cosmic dawn.

These timely observations coincide with JWST's groundbreaking ability to directly observe globular cluster (GC) formation, and provide a bridge between the old stellar populations that we see in GCs today and the young clusters they are thought to form from. Our key science goals are to: (i) characterize the extreme ISM conditions in the clusters, (ii) determine whether they form in a single burst or show extended star formation, (iii) investigate the potential evidence for an excess of massive stars expected in early star formation using Nitrogen abundances, and (iv) evaluate the possibility of a clumpy CGM by spatially resolving the Lyman-break. These observations will significantly advance our understanding of the birth of the first stellar systems in a low-mass galaxy at extreme redshifts, probe the connection between massive star clusters in young galaxies and future globular clusters, and provide synergies with upcoming ALMA measurements.

OBSERVING DESCRIPTION

We will use NIRSpec/IFU with CLEAR/PRISM to observe the galaxy for 41.5 hours. This will provide a continuum SNR per pixel of $\sim 5-15$ for the star clusters and $\sim 2-3$ at $2\ \mu\text{m}$ for the diffuse galaxy body. The entire FS fits diagonally within the IFU field of view (FoV). We will use a 4-point dither pattern centered at four locations along the galaxy. This 16-point dither ensures robust background spectra free from contamination by the nearby bright cluster galaxy, while covering all star clusters of interest. The central brightest clusters, crucial for ISM studies, will remain in the FoV for all 16 pointings, achieving an SNR of 20 for the continuum and 40-80 for emission lines. The outermost clusters, with the

JWST Proposal 7049 (Created: Monday, June 22, 2026, 6:00:11PM Eastern Standard Time) - Overview

lowest magnification, will be in the FoV for half the pointings, allowing us to determine their stellar properties. We will use the NRSIRS2 readout pattern, which reduces correlated $1/f$ noise, crucial for sky background acquisition. Each pointing will include 32 groups and 4 integrations. To avoid leakage through the MSA, we request a V3PA constraint of 191 to 216 and 296 to 112 degrees, ensuring that bright stars and cluster galaxies are behind the MSA's solid metal plates. Blind pointing target acquisition, accurate to $0.1''$, will position the galaxy near the center of the $3'' \times 3''$ FoV.

Proposal 7049 - Targets - A Unique Opportunity to Probe Lensing-Magnified Star Clusters in a Low-Mass Galaxy at $z_{\text{spec}}=8.3$ with J...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Firefly	RA: 14 23 46.1747 (215.9423946d) Dec: +24 04 10.74 (24.06965d) Equinox: J2000	Epoch of Position: 2000	
Comments: Category=Galaxy Description=[High-redshift galaxies, Protogalaxies] Extended=YES					

Proposal 7049 - Observation 1 - A Unique Opportunity to Probe Lensing-Magnified Star Clusters in a Low-Mass Galaxy at $z_{\text{spec}}=8.3...$

Observation	Proposal 7049, Observation 1 Mon Jun 22 23:00:12 GMT 2026											
	Diagnostic Status: Warning Observing Template: NIRSspec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous			
	(1)	Firefly	RA: 14 23 46.1747 (215.9423946d) Dec: +24 04 10.74 (24.06965d) Equinox: J2000				Epoch of Position: 2000					
Template	<i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[High-redshift galaxies, Protogalaxies]</i> <i>Extended=YES</i>											
	TA Method						HFF Readout Mode					
Dithers	NONE						false					
	#	Dither Type		Size		Starting Point		Number of Points		Points		
Spectral Elements	1	CYCLING		LARGE		2		32				
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
Special Requirements	1	PRISM/CLEAR	NRSIRS2	21	3	false	true	NONE	32	96	148456.545	
	Group Visits within 53.0 Days Aperture PA Range 74.97164917 to 250.97164917 Degrees (V3 296.0 to 112.0) Aperture PA Range 329.97164917 to 354.97164917 Degrees (V3 191.0 to 216.0) Visits Same PA											